Course- B. Tech
Course Code- CSET343
Year- 4th Year

Type- Specialization Elective
Course Name- AI in Healthcare
Semester- VII

Lab Assignment – 8

Objective: To analyze and process real-world healthcare-related text data using NLP techniques such as topic modeling, thereby understanding how language data can reveal insights about public health discussions and medical trends.

Dataset: https://github.com/WuraolaOyewusi/Health-Related-Headlines-Datasets-for-Natural-Language-Processing

This dataset contains **health-related headlines, titles, and teasers** from various **news sources, journals, and web articles**.

Task 1: Data Preprocessing and Cleaning

- Load the dataset (CSV/JSON format).
- Perform:
  - Tokenization
  - Stop-word removal
  - Lemmatization
  - Lowercasing and punctuation removal
- Display a few cleaned text samples.

Task 2: Exploratory Text Analysis & Word Visualization

- Generate:
  - Word frequency distribution
  - Word Cloud showing the most common health-related terms.
- Plot top 20 frequent words using matplotlib or seaborn.

Task 3: Topic Modelling using LDA (Latent Dirichlet Allocation)

- Use the cleaned text corpus to identify dominant health-related topics.
- Display top 5 words per topic and visualize the results using pyLDAvis.